F78AB – Actuarial and Financial Mathematics
Computer Lab 1

In this lab we will use Excel to redo some of the questions from the lectures and tutorials. Below are some good practices to follow as you work in Excel.

- Your calculations should be broken down into small steps. This makes your calculations easier to check and follow.
- As far as reasonably possible, you should assign an Excel name to each constant used in Excel. This makes your spreadsheet easier to follow and modify if the constants change.
- Avoid Hard Coding as much as possible.
- Save your work frequently.
- It is a good idea to save your workbook (Excel file) with changes to the version number (in case you want to go back to an earlier version).

For all the questions you should do the calculations from the individual cashflows and NOT use the annuity formulae. You could use the annuity formulae to check that you get the same answer.

Question 1:

Use Excel to do the calculations for **example 3.2.1** from the notes.
EXTRA: Use Excel to do the calculations for **example 3.3.1** from the notes.

Question 2:

Use Excel to do the calculations for **example 3.6.1** from the notes.

Question 3:

Use Excel to do the calculations for **example 3.7.2** from the notes.
Do this in the following two ways:

a) Use the actual cashflows.
b) Split the cashflows into a level annuity and an increasing annuity i.e. $9000a_{\overline{20}|} + 1000(Ia)_{\overline{20}|}$.

EXTRA: Use Excel to do the calculations for **example 3.7.3** from the notes.

Question 4:

Use Excel to do the calculations for **example 3.7.4** from the notes.

Question 5:

Do the calculations for **Tutorial 2 Question 13** in Excel using

a) the effective rate of interest per year, and
b) the effective rate of interest per quarter year.